# Object Storage Services
# Amazon S3
# Tutorial

## Topics

1. Using Amazon S3 via the AWS SDK for Python

**Prerequisite**

- Complete this tutorial using the AWS Academy Learner Lab dedicated for our module[1].
- Note that if you are completing this tutorial using the AWS Academy Learner Lab, there is a restriction on the **AWS Regions** that can be used, namely all service access is limited to the **us-east-1** and **us-west-2** AWS Regions unless mentioned otherwise.
- This tutorial assumes that you are using the AWS Cloud9 Environment set up for developing in Python[2]. Otherwise please complete the section *Setting up Cloud9 for Cloud Application Development in Python* of the Tutorial on AWS Cloud9 for Python.

## Working with Amazon S3

Amazon Simple Storage Service (Amazon S3)[3] is an object storage service that provides scalability, data availability, security and performance. It enables us to store large amounts of unstructured data.

[1] This assumes that you have accepted the invitation to join the AWS Academy Learner Lab received via your NCI email account.

[2] https://docs.aws.amazon.com/cloud9/latest/user-guide/sample-python.html

[3] https://aws.amazon.com/s3/

The aim of this tutorial is to learn how to programmatically use Amazon S3. More specifically, at the end of this tutorial you should know to perform the following tasks:

- Create an Amazon S3 bucket
- List existing buckets
- Upload files to a bucket
- List all the objects in a given bucket
- Download an S3 object to a file
- Delete object(s)
- Delete a bucket

Recall that the AWS SDKs provide two levels of APIs for the AWS services, namely Client (or "low-level") APIs and Resource APIs.

In this tutorial we will use the S3 Client API which is a low-level client representing the Amazon Simple Storage Service (S3). Please consult the documentation of the S3 Client at https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/s3.html#client

**Note** The folder *s3_demos* available on the module's Moodle page shows examples of how to use Amazon S3 programmatically via the S3 Client API. Explanations are provided during class.

**Create an Amazon S3 bucket**
boto3 S3 Client method

```
create_bucket()
```

By default, a bucket is created in the US East (N. Virginia) Region (i.e., us-east-1). When we create a bucket we can specify the region where the bucket is created (e.g., US West (Oregon) Region (i.e., us-west-2)). For further details please consult the boto3 S3 Client create_bucket() method.

***Recall*** *The name of an S3 bucket must be unique across all regions!*

**List existing buckets**
boto3 S3 Client method

```
list_buckets()
```

Returns a list of all buckets owned by the authenticated sender of the request. For further details please consult the boto3 S3 Client list_buckets() method.

**Upload files to a bucket**
boto3 S3 Client method

```
upload_file()
```

Upload a file to an S3 object. For further details please consult the boto3 S3 Client upload_file() method.

If you have a file-like object in binary format and you would like to perform a managed upload which may perform a multipart upload in multiple threads (if needed), please consult the boto3 S3 Client upload_fileobj() method.

**Delete object(s)**
boto3 S3 Client methods

```
delete_object()
delete_objects()
```

The *delete_objects()* method enables to delete multiple objects from a bucket using a single HTTP request. For further details please consult the boto3 S3 Client delete_object() and delete_objects() methods.

**Delete a bucket**
boto3 S3 Client method

```
delete_bucket()
```

Deletes an S3 bucket. Note that all the objects (including all object versions and delete markers) in the bucket must be deleted before the bucket itself can be deleted. For further details please consult the boto3 S3 Client delete_bucket() method.

**Running the sample Python program**

To run the sample Python program *s3_ demo.py*[4] run at the terminal the following command

```
python s3_demo.py <bucket-name> --file_name <file> --object_key <key>
```

where

- `<bucket-name>` should be replaced with the name of your bucket – for example, in addition to the name of the bucket add your initials and date to ensure bucket name *uniqueness!*
- `--file_name`

  is an optional argument needed when uploading a file to a bucket
- `--object_key`

  is an optional argument needed when performing operations such as uploading a file to a bucket or deleting an object

For example,

```
python s3_demo.py my-photos-aec-1103 --file_name crocuses.jpg --object_key crocus
```

**TASKS**

1. **List all the objects in a given bucket**

   **TASK** Using the previous examples from *s3_ demo.py* as an introduction to working with S3 via the S3 client APIs, implement a method to list/display all the objects from a given S3 bucket.

   *Hint* First, identify the S3 client method that provides this operation.

2. **Download an S3 object to a file**

   **TASK** Using the previous examples from *s3_ demo.py* as an introduction to working with S3 via the S3 client APIs, implement a method which enables us to download an S3 object to a file.

   *Hint* First, identify the S3 client method that provides this operation.

[4] *s3_ demo.py* is available on the module's Moodle page

3. **Enable versioning on a bucket**

   **TASK** Using the previous examples from *s3_ demo.py* as an introduction to working with S3 via the S3 client APIs, implement a method to enable bucket-versioning on a given bucket, and then upload 3 different version of the same file to that bucket.

## Resources

- Amazon S3 https://aws.amazon.com/s3/
- Boto3 Amazon S3 Documentation https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/s3.html
- Boto3 Amazon S3 Client https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/s3.html#client
- AWS Cloud9 https://docs.aws.amazon.com/cloud9/latest/user-guide/welcome.html
- Python Tutorial for AWS Cloud9 https://docs.aws.amazon.com/cloud9/latest/user-guide/sample-python.html